

INGREDION (THAILAND) CO., LTD. - KALASIN STARCH PLANT

PROJECT JET COOKER

Interconnection Cable Schedule

Complete Multi-Core & Trunk Routing Between MCC / CA1 Suite and Field Junction Boxes

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Engineering Contractor: AEC Industrial Engineering Co., Ltd.

Overall Cable Routing & Signal Segregation Architecture

Physical pathways from MCC Room to Infeed, Jet Cooker, Spray Dryer, Packing Tower, and Slurry Building

Substation to Spine (TR-01)

Main heavy-duty hot-dip galvanized ladder tray (600x100mm) running from MCC room basement trench across overhead pipe rack.

Spray Dryer Riser (TR-SD)

Vertical cable riser traversing 1st to 8th floors. Segregated trays for Non-IS Control, Analog Shielded, and Ex-i Blue Trays.

Packing Tower Riser (TR-PT)

Heavy-duty ladder tray for 2nd and 8th floor junction boxes. Blue-coated Ex-i dedicated cable tray with 300mm air gap separation.

Slurry Bldg Fiber Trunk (TR-EXT)

Overhead pipe rack route to remote building housing RIO-200. 6-core steel wire armoured fiber optic DLR ring backbone.

ENGINEERING SEGREGATION & INSTALLATION RULES (IEC 60079-14 & IEEE 518)

1. Intrinsically Safe (Ex-i) Cables: Distinctive BLUE outer sheath (RAL 5015). Maintain ≥ 300 mm clearance or solid grounded divider from power/control.
2. Shield Grounding: All overall and individual pair screens grounded at MCC Cabinet single-point clean earth bar ONLY. Field ends insulated and floating.
3. Spare Capacity: All multicore control cables sized with 20% to 35% installed spare conductors for future process expansion.
4. Cable Glands: Nickel-plated brass double-compression EMC cable glands (IP66) on standard JBs; Ex-e/Ex-i certified blue-ring glands on Ex-i JBs.
5. Testing & Verification: Insulation resistance test (Megger 500VDC > 20 MOhm) and loop continuity resistance test required before energization.

Panel M1 Interconnection Cable Schedule

Master Controller Bay (=CA1+MCP-M1) serving Infeed (JB-401), Spray Dryer 1F (JB-601), and Slurry (RIO-200)

Cable Tag	From	To (JB)	Location	Signal Function	Cable Specification	Tray	Length
CBL-M1-JB401-01	Panel M1	JB-401	Infeed 2nd Flr	Digital Inputs (DI)	CVV-S 48C x 1.5 mm2	TR-IN-01	55 m
CBL-M1-JB401-02	Panel M1	JB-401	Infeed 2nd Flr	DI Extension & Spares	CVV-S 30C x 1.5 mm2	TR-IN-01	55 m
CBL-M1-JB401-03	Panel M1	JB-401	Infeed 2nd Flr	Digital Outputs (DO)	CVV-S 40C x 1.5 mm2	TR-IN-01	55 m
CBL-M1-JB401-04	Panel M1	JB-401	Infeed 2nd Flr	Analog Inputs (AI)	IS/OS/PVC 16P x 1.0 mm2	TR-IN-01	55 m
CBL-M1-JB601-01	Panel M1	JB-601	Spray Dryer 1st Flr	Digital Inputs (DI)	CVV-S 60C x 1.5 mm2	TR-SD-01	70 m
CBL-M1-JB601-02	Panel M1	JB-601	Spray Dryer 1st Flr	Digital Outputs (DO)	CVV-S 24C x 1.5 mm2	TR-SD-01	70 m
CBL-M1-JB601-03	Panel M1	JB-601	Spray Dryer 1st Flr	Analog Inputs (AI)	IS/OS/PVC 16P x 1.0 mm2	TR-SD-01	70 m
CBL-M1-RIO200-FO	Panel M1	RIO-200	Slurry Out Bldg 2F	Redundant DLR Ethernet	6-Core Armoured SM Fiber (OS2)	TR-EXT-01	165 m
CBL-M1-RIO200-PWR	Panel M1	RIO-200	Slurry Out Bldg 2F	230VAC UPS Power	XLPE/SWA/PVC 3C x 4.0 mm2	TR-EXT-01	165 m

Panel M2 Interconnection Cable Schedule

High-Density I/O Bay (=CA1+MCP-M2) serving Jet Cooker (JB-402), Spray Dryer 3F (JB-602), and Spray Dryer 7F (JB-607)

Cable Tag	From	To (JB)	Location	Signal Function	Cable Specification	Tray	Length
CBL-M2-JB402-01	Panel M2	JB-402	Jet Cooker 2nd Flr	Digital Inputs (DI)	CVV-S 40C x 1.5 mm2	TR-JC-01	45 m
CBL-M2-JB402-02	Panel M2	JB-402	Jet Cooker 2nd Flr	Digital Outputs (DO)	CVV-S 24C x 1.5 mm2	TR-JC-01	45 m
CBL-M2-JB402-03	Panel M2	JB-402	Jet Cooker 2nd Flr	Analog Inputs & AO	IS/OS/PVC 20P x 1.0 mm2	TR-JC-01	45 m
CBL-M2-JB602-01	Panel M2	JB-602	Spray Dryer 3rd Flr	Digital Inputs (DI)	CVV-S 40C x 1.5 mm2	TR-SD-02	85 m
CBL-M2-JB602-02	Panel M2	JB-602	Spray Dryer 3rd Flr	Digital Outputs (DO)	CVV-S 24C x 1.5 mm2	TR-SD-02	85 m
CBL-M2-JB602-03	Panel M2	JB-602	Spray Dryer 3rd Flr	Analog Inputs & AO	IS/OS/PVC 36P x 1.0 mm2	TR-SD-02	85 m
CBL-M2-JB607-01	Panel M2	JB-607	Spray Dryer 7th Flr	Digital Inputs (DI) Pt 1	CVV-S 60C x 1.5 mm2	TR-SD-03	125 m
CBL-M2-JB607-02	Panel M2	JB-607	Spray Dryer 7th Flr	Digital Inputs (DI) Pt 2	CVV-S 60C x 1.5 mm2	TR-SD-03	125 m
CBL-M2-JB607-03	Panel M2	JB-607	Spray Dryer 7th Flr	Digital Outputs (DO) Pt 1	CVV-S 48C x 1.5 mm2	TR-SD-03	125 m
CBL-M2-JB607-04	Panel M2	JB-607	Spray Dryer 7th Flr	Digital Outputs (DO) Pt 2	CVV-S 30C x 1.5 mm2	TR-SD-03	125 m
CBL-M2-JB607-05	Panel M2	JB-607	Spray Dryer 7th Flr	Analog Inputs & AO	IS/OS/PVC 40P x 1.0 mm2	TR-SD-03	125 m

Panel M3 Intrinsically Safe (Ex-i) Cable Schedule

Ex-i Hazardous Area Bay 1 (=CA1+MCP-M3) with Pepperl+Fuchs Isolators serving IS-JB-603, IS-JB-618, and JB-618

Cable Tag	From	To (JB)	Location	Signal Function	Cable Specification	Tray	Length
CBL-M3-ISJB603-01	Panel M3	IS-JB-603	Spray Dryer 3rd Flr	Ex-i Digital Inputs (DI)	CVV-S (IS) 24C x 1.5 mm2 [BLUE]	TR-SD-IS	85 m
CBL-M3-ISJB603-02	Panel M3	IS-JB-603	Spray Dryer 3rd Flr	Ex-i Analog Inputs (AI)	IS/OS/PVC (IS) 16P x 1.0 mm2 [BLUE]	TR-SD-IS	85 m
CBL-M3-ISJB618-01	Panel M3	IS-JB-618	Packing Tower 8th Flr	Ex-i Digital Inputs (DI)	CVV-S (IS) 40C x 1.5 mm2 [BLUE]	TR-PT-IS	140 m
CBL-M3-ISJB618-02	Panel M3	IS-JB-618	Packing Tower 8th Flr	Ex-i Analog Inputs (AI)	IS/OS/PVC (IS) 16P x 1.0 mm2 [BLUE]	TR-PT-IS	140 m
CBL-M3-JB618-01	Panel M3	JB-618	Packing Tower 8th Flr	Non-IS DI & AI	CVV-S 24C x 1.5 mm2	TR-PT-01	140 m
CBL-M3-JB618-02	Panel M3	JB-618	Packing Tower 8th Flr	Non-IS Digital Outputs	CVV-S 40C x 1.5 mm2	TR-PT-01	140 m

Panel M4 Intrinsically Safe (Ex-i) Cable Schedule

Ex-i Hazardous Area Bay 2 (=CA1+MCP-M4) serving IS-JB-608, IS-JB-612, JB-606, JB-608, and JB-612

Cable Tag	From	To (JB)	Location	Signal Function	Cable Specification	Tray	Length
CBL-M4-ISJB608-01	Panel M4	IS-JB-608	Spray Dryer 8th Flr	Ex-i DI & DO Signals	CVV-S (IS) 30C x 1.5 mm2 [BLUE]	TR-SD-IS	135 m
CBL-M4-ISJB608-02	Panel M4	IS-JB-608	Spray Dryer 8th Flr	Ex-i Analog Inputs (AI)	IS/OS/PVC (IS) 10P x 1.0 mm2 [BLUE]	TR-SD-IS	135 m
CBL-M4-ISJB612-01	Panel M4	IS-JB-612	Packing Tower 2nd Flr	Ex-i Digital Inputs (DI)	CVV-S (IS) 40C x 1.5 mm2 [BLUE]	TR-PT-IS	65 m
CBL-M4-ISJB612-02	Panel M4	IS-JB-612	Packing Tower 2nd Flr	Ex-i Analog Inputs (AI)	IS/OS/PVC (IS) 12P x 1.0 mm2 [BLUE]	TR-PT-IS	65 m
CBL-M4-JB606-01	Panel M4	JB-606	Spray Dryer 6th Flr	Non-IS DI, DO & AI	CVV-S 30C x 1.5 mm2	TR-SD-02	115 m
CBL-M4-JB608-01	Panel M4	JB-608	Spray Dryer 8th Flr	Non-IS Analog Inputs	IS/OS/PVC 8P x 1.0 mm2	TR-SD-03	135 m
CBL-M4-JB612-01	Panel M4	JB-612	Packing Tower 2nd Flr	Non-IS Digital Inputs	CVV-S 40C x 1.5 mm2	TR-PT-01	65 m
CBL-M4-JB612-02	Panel M4	JB-612	Packing Tower 2nd Flr	Non-IS DO & AI	CVV-S 24C x 1.5 + 10P x 1.0	TR-PT-01	65 m

MCC Switchgear & Remote I/O Trunk Schedule

Hardwired control interlocks to Motor Control Center switchgear and Armoured Fiber link to RIO-200

Cable Tag	From	To (JB)	Location	Signal Function	Cable Specification	Tray	Length
CBL-M1-RIO200-FO	Panel M1	RIO-200	Slurry Out Bldg 2F	Redundant DLR Ethernet	6-Core Armoured SM Fiber (OS2)	TR-EXT-01	165 m
CBL-M1-RIO200-PWR	Panel M1	RIO-200	Slurry Out Bldg 2F	230VAC UPS Power	XLPE/SWA/PVC 3C x 4.0 mm2	TR-EXT-01	165 m
CBL-M1-MCC-01	Panel M1	MCC Lineup	MCC Room	Motor Feedback Signals	2x CVV-S 48C + 36C x 1.5 mm2	TR-MCC-01	25 m
CBL-M1-MCC-02	Panel M1	MCC Lineup	MCC Room	Motor Command Signals	2x CVV-S 48C + 24C x 1.5 mm2	TR-MCC-01	25 m
CBL-M1-MCC-NET	Panel M1	MCC Lineup	MCC Room	Smart Overloads (E300)	Armoured Industrial Cat6A DLR	TR-MCC-01	30 m

Master Cable Bill of Quantities (BOQ) & Drum Schedule

Consolidated procurement quantities with 10% installation slack and drum packaging

Item No.	Cable Specification & Construction	Signal Category	Runs	Total Length	Drum Packaging
CBL-01	CVV-S 48C x 1.5 mm2	Control	2	198 m	1x 198m Drum
CBL-02	CVV-S 30C x 1.5 mm2	Control	3	324 m	1x 324m Drum
CBL-03	CVV-S 40C x 1.5 mm2	Control	5	429 m	1x 429m Drum
CBL-04	IS/OS/PVC 16P x 1.0 mm2	Analog Shielded	2	137 m	1x 137m Drum
CBL-05	CVV-S 60C x 1.5 mm2	Control	3	352 m	1x 352m Drum
CBL-06	CVV-S 24C x 1.5 mm2	Control	4	374 m	1x 374m Drum
CBL-07	6-Core Armoured SM Fiber (OS2)	Fiber Optic	1	181 m	1x 181m Drum
CBL-08	XLPE/SWA/PVC 3C x 4.0 mm2	Power	1	181 m	1x 181m Drum
CBL-09	IS/OS/PVC 20P x 1.0 mm2	Analog Shielded	1	49 m	1x 49m Drum
CBL-10	IS/OS/PVC 36P x 1.0 mm2	Analog Shielded	1	93 m	1x 93m Drum
CBL-11	IS/OS/PVC 40P x 1.0 mm2	Analog Shielded	1	137 m	1x 137m Drum
CBL-12	CVV-S (IS) 24C x 1.5 mm2 [BLUE]	Ex-i Intrinsically Safe	1	93 m	1x 93m Drum
CBL-13	IS/OS/PVC (IS) 16P x 1.0 mm2 [BLUE]	Ex-i Intrinsically Safe	2	247 m	1x 247m Drum
CBL-14	CVV-S (IS) 40C x 1.5 mm2 [BLUE]	Ex-i Intrinsically Safe	2	225 m	1x 225m Drum
CBL-15	CVV-S (IS) 30C x 1.5 mm2 [BLUE]	Ex-i Intrinsically Safe	1	148 m	1x 148m Drum
CBL-16	IS/OS/PVC (IS) 10P x 1.0 mm2 [BLUE]	Ex-i Intrinsically Safe	1	148 m	1x 148m Drum
CBL-17	IS/OS/PVC (IS) 12P x 1.0 mm2 [BLUE]	Ex-i Intrinsically Safe	1	71 m	1x 71m Drum
CBL-18	IS/OS/PVC 8P x 1.0 mm2	Analog Shielded	1	148 m	1x 148m Drum
CBL-19	CVV-S 24C x 1.5 + 10P x 1.0	Control & Analog	1	71 m	1x 71m Drum
CBL-20	2x CVV-S 48C + 36C x 1.5 mm2	Control	1	27 m	1x 27m Drum
CBL-21	2x CVV-S 48C + 24C x 1.5 mm2	Control	1	27 m	1x 27m Drum